BUILDING A USB RC METER

Episode 2 Worksheet

The Transistor as a Constant-Current Source

Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.

Learning objectives checked

- Explain why a meter needs a constant current, not just a resistor
- Identify the cutoff, active and saturation regions of a BJT
- Design a PNP current source step by step (R1, R2, R3, D1)
- Predict and measure the compliance limit, and explain the temperature diode

Questions

1. A PNP current source has V_{CC} = 5 V, base held at 3.6 V, V_{EB} = 0.65 V and R_E = 510 Ω. Find the emitter current.

2. Using β = 300, estimate the base current for 1.5 mA collector current and compare it with the ≈1.1 mA divider current.

3. Redesign R3 for a 1.0 mA stage, assuming 0.8 V across R3. Pick an E24 value.

4. With V_E = 4.2 V and $V_{EC,sat}$ = 0.2 V, what's the largest resistor Stage 2 (160 µA) can measure?

5. Without the diode, V_{RE} rises 2 mV/°C. For V_{RE} = 0.8 V, what's the current change for +15 °C?

6. Why does Stage 4 (510 kΩ) give 1.82 µA instead of the 1.57 µA predicted from 0.8 V / 510 kΩ?

7. Name the three BJT regions and which one a current source must operate in.

8. Why was a PNP chosen instead of an NPN for this meter?

Answers

1. $V_E = 3.6 + 0.65 = 4.25$ V; $V_{RE} = 5 - 4.25 = 0.75$ V; $I_E = 0.75 / 510 \approx 1.47$ mA.
2. $I_B = 1.5$ mA / 300 = 5 µA, about 0.5 % of the divider current, so the divider is stiff.
3. R3 = 0.8 / 1.0 mA = 800 Ω → 820 Ω (≈0.98 mA).
4. (4.2 − 0.2) / 160 µA = 25 kΩ (the thesis used 23.7 kΩ at the 3.8 V ADC limit).
5. ΔV = 30 mV → 30 / 800 = +3.75 %.
6. At 1.8 µA the base-emitter voltage is ≈3 decades × 60 mV ≈ 0.18 V lower than at 1.5 mA, so $V_{RE} \approx 0.93$ V → 1.82 µA.
7. Cutoff, active (forward-active), saturation. A current source must be in the ACTIVE region.
8. So the resistor under test can have one end grounded and the ADC can measure its voltage directly with respect to ground.

Going further

BC559 datasheet (NXP, via DigiKey): PNP, 100 mA, 500 mW: https://media.digikey.com/pdf/Data%20Sheets/NXP%20PDFs/BC559.pdf

BC559 datasheet (onsemi): 625 mW version: https://www.onsemi.com/download/data-sheet/pdf/bc559-d.pdf

1N4148 datasheet (Vishay): https://www.vishay.com/docs/81857/1n4148.pdf

Horowitz & Hill, The Art of Electronics (3rd ed.), ch. 2: current sources: https://artofelectronics.net/

All About Circuits: BJT current mirrors and sources: https://www.allaboutcircuits.com/textbook/semiconductors/chpt-4/current-mirrors/

Falstad circuit simulator (in-browser): https://www.falstad.com/circuit/